

Composition to Verify Inverses

Algebra II | Unit 1 | Day 13 | TEK A2.2.D

Objective: I can compose two functions and use the results $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$ to prove whether they are inverses of each other.

Vocabulary

Composition of functions — putting one function _____ another and evaluating: $f(g(x))$, read "f of g of x." Work from the _____.

Part 1 — Warm Up on Composition

Let $f(x) = 3x - 5$ and $g(x) = (x + 5)/3$. Work inside first.

$$g(2) = \underline{\hspace{2cm}} \quad f(4) = \underline{\hspace{2cm}} \quad f(g(4)) = f(\underline{\hspace{2cm}}) = \underline{\hspace{2cm}}$$

$$g(f(4)) = g(\underline{\hspace{2cm}}) = \underline{\hspace{2cm}} \quad \text{Notice both came back to } \underline{\hspace{2cm}}, \text{ the number we started with.}$$

Part 2 — Prove It with Algebra

Example 1. $f(x) = 3x - 5$ and $g(x) = (x + 5)/3$

$$f(g(x)) = 3 \cdot \underline{\hspace{2cm}} - 5 = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$g(f(x)) = (\underline{\hspace{2cm}} + 5)/3 = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

Both give x , so f and g _____ inverses.

You Try 1. $f(x) = 2x + 3$ and $g(x) = (x - 3)/2$

$$f(g(x)) = \underline{\hspace{2cm}} \quad g(f(x)) = \underline{\hspace{2cm}} \quad \text{Inverses? } \underline{\hspace{2cm}}$$

You Try 2. $f(x) = x + 6$ and $g(x) = x - 6$

$$f(g(x)) = \underline{\hspace{2cm}} \quad g(f(x)) = \underline{\hspace{2cm}} \quad \text{Inverses? } \underline{\hspace{2cm}}$$

Part 3 — A Pair That Is NOT Inverse

Test $f(x) = 2x + 1$ and $g(x) = 2x - 1$. Do not assume - compute.

$$f(g(x)) = 2(\underline{\hspace{2cm}}) + 1 = \underline{\hspace{2cm}} \quad \text{Does this equal } x? \underline{\hspace{2cm}}$$

So they are _____ inverses. What is the actual inverse of $f(x) = 2x + 1$? $f^{-1}(x) = \underline{\hspace{2cm}}$

Lesson: changing a sign is _____ the same as undoing the function. You must undo every step, in reverse order.

Part 4 — Error Analysis

A student is given $f(x) = 2(x - 3)^2 + 4$ and claims $f^{-1}(x) = \sqrt{x - 3} + 4$.

Test it: is $f(f^{-1}(x)) = x$? _____ So the claim is _____.

Find the mistake. Undo the steps in reverse: subtract 4, divide by 2, take the square root, then add 3.

Correct inverse: $f^{-1}(x) = \underline{\hspace{2cm}}$

Why does a $\pm$ show up here? _____

What restriction would make the inverse a function? _____

Part 5 — Self-Inverse Functions

A few functions are their own inverse - composing them with themselves gives x .

1. $f(x) = x$ $f(f(x)) =$ _____ Self-inverse? _____
2. $f(x) = -x$ $f(f(x)) =$ _____ Self-inverse? _____
3. $f(x) = 1/x$ $f(f(x)) =$ _____ Self-inverse? _____

What do their graphs share? _____

Practice

Decide whether each pair is inverse. Show BOTH compositions - one direction is not proof.

1. $f(x) = 5x$ and $g(x) = x/5$

$f(g(x)) =$ _____ $g(f(x)) =$ _____ Inverses? _____

2. $f(x) = x - 3$ and $g(x) = 3 - x$

$f(g(x)) =$ _____ Inverses? _____ Correct inverse of f : _____

3. $f(x) = (x + 1)/2$ and $g(x) = 2x - 1$

$f(g(x)) =$ _____ $g(f(x)) =$ _____ Inverses? _____

4. $f(x) = x^2$ and $g(x) = \sqrt{x}$. Test with $x = -4$ as well as $x = 4$.

Does it work for every real x ? _____ Restriction needed: _____

5. Write the inverse of $f(x) = 7x + 2$, then verify it by composition.

$f^{-1}(x) =$ _____

Exit Ticket

Decide whether $f(x) = 4x - 8$ and $g(x) = (x + 8)/4$ are inverses. Show both compositions.

$f(g(x)) =$ _____ $g(f(x)) =$ _____ Inverses? _____

Why is checking only one direction not enough? _____